

Quantum-Enabled Grid Expansion Planning for Distribution Networks

hard instances · a NISQ algorithm · a certified line set · a real-time contingency spectrum

Seven guideline sections · every prior-work number carries a receipt

Public repository: <https://github.com/sharadbachani-oss/merlin-quantum-eon-gic2026>

Summary

−98.1%

thermal congestion on the certified plan (77.125 → 1.440), 220 fewer overloaded (line, scenario, contingency) triples, 286 voltage violations cleared, capex 11.42 — the exact minimum, recovered on Dirac-3 and by QAOA on IBM Heron

10.5%

better than the proof-grade MILP incumbent at its 901 s timeout on the 20-variable hard rung: device basin-finding in seconds plus a one-flip polish reaches 22,057, the best value known for the instance

29%

storage undersize that the imaginary-time continuation route produces on one microgrid decision, avoided by the real-time leftover spectrum — a planning input whose scalable classical route is ill-posed at any runtime

Why the numbers can be trusted

One network-physics referee (DC-PTDF thermal congestion plus LinDistFlow voltage) grades every solver. Hardness is a measured convergence law, not a timeout: the 26-variable proof gap closes to 22.6% after 2.6 h and extrapolates to 17–21% unproven after a full afternoon, with exhaustive certification at 87 CPU-days. Every hardware number carries a cloud job ID and a frozen pre-registration; a credential-free audit replays every headline in about three minutes.

This is a Phase-1 concept proposal in the seven guideline sections. The Phase-2 PoC brings Gurobi/CPLEX proof gaps, the full-k four-plan leftover series on hardware, and E.ON's anonymised MV/LV subgrids.

Concept Proposal

1. Problem framing

Distribution expansion is a build-or-no-build decision over candidate corridors, coupled to network physics across many operating scenarios and N–1 contingencies. The statement's own diagnosis is that the combinatorial explosion forces planners toward conservative plans because they cannot evaluate enough of them. We follow the statement's named framework [5] in its order: identify instances that proof-grade classical solvers cannot certify in reasonable time, then shape the quantum approach to those instances.

Two things make a credible quantum advantage here. First, the hard instances are real: on our 26-binary instance the proof gap is a measured convergence law (33.8% at 480 s → 22.6% after 2.6 h), extrapolating to 17–21% still unproven after a full afternoon and 3–52 days to a certificate; exhaustive certification costs 87 CPU-days. Second, and more decisive, what a DSO actually needs to size storage or reinforcement is not a static objective but the **rhythm with which the existing slack corridors ring after a contingency** — a real-time dynamical spectrum $S(\omega)$ of the reinforced network. The scalable classical route to that object, imaginary-time evolution plus analytic continuation, is mathematically ill-posed at any runtime: on our 20-qubit instance two spectra fit the same imaginary-time data and neither recovers the real-time line (0.270 / 0.253 vs 0.222). A quantum processor evolves on the real-time axis natively; the spectrum is a Fourier transform with no continuation step. That is an advantage of route, it holds at 20 qubits today, and it is the class of computation every 2025–26 verified quantum-advantage candidate occupies.

2. Technical approach

Paradigm. Hybrid. (a) Gate-model real-time dynamics on IBM Heron: the candidate graph embedded as an interacting-lattice operator $H = \mu^* \cdot D - A \blacksquare$ (one candidate corridor = one qubit pair; existing corridors always in the Hamiltonian; built closures enter as Hamiltonian support), exact per-rung ground-state preparation, Trotterised N–1 quench, leftover series $L(k)$ on the slack pairs and its spectrum. (b) Variational: 10-qubit QAOA (depth 333) on the same instance. (c) Photonic entropy computing (QCi Dirac-3) holding the full degree- ≤ 5 expansion polynomial for the hard instances. (d) Classical: one network-physics referee (DC-PTDF thermal congestion + LinDistFlow voltage) that grades every plan from every solver identically.

Instances. IEEE 14-bus (pandapower distribution) and synthetic grids matched to its graph properties; five instances, 6 → 26 binaries, 2–6 scenarios, N–1 on the hard rungs.

Hardware discipline. Equal-structure parameterised templates transpiled once with fixed layout; visibility-inverted readout (per-rung visibility 0.93–0.98); null (no-quench) and detuned controls in every job; cross-device replication (ibm_kingston → ibm_marrakesh). Compilation is native heavy-hex with zero SWAP overhead; a 64-rung (128-qubit) series is the same compilation, tiled.

3. Feasibility and resource requirements

resource	status
Instances and referee	in package; python <code>verify.py</code> (numpy only) replays every headline number from archived counts in ~3 min
Proof-grade classical	HIGHS (open, MILP with proof gap) — receipted; Gurobi/CPLEX replication of the eon57 rows is PoC step 1 (licence required)
Gate-model hardware	IBM Heron via Startup Program; the $k = 0..8$ four-plan leftover series is 54 logical circuits, 442,368 shots; the 64-rung series is the same emitter tiled
Photonic	QCi Dirac-3, unmetered allocation
Data	open test systems; Phase 2 targets E.ON's anonymised MV/LV subgrids
Classical compute	32-core workstation for enumeration, MPS comparators and multi-start search

Assumptions: DC-PTDF + LinDistFlow referee (penalised DC objective, not full AC feasibility — AC recourse is charged as a classical solve per accepted plan in Phase 2). Constraint: hardware ranking at 20 qubits is currently two-point; the full-k series is the PoC's first flight.

4. Expected impact

Hard instances that scale poorly (statement §4.1, receipted):

instance	build vars	scenarios	N-1	MILP vars	HiGHS	outcome
eon14_easy	6	2	no	96	0.28 s	proven optimal
eon14_hard	10	4	yes	4,250	4.5 s	proven — the NISQ instance
eon30_hard	14	4	yes	17,614	183 s	proven optimal
eon57_util	20	4	yes	59,116	900 s	timeout, 28.25% gap
eon57_deep	26	6	yes	94,646	1,801 s	timeout, 33.8% gap → 22.6% at 2.6 h; 3–52 days to certificate

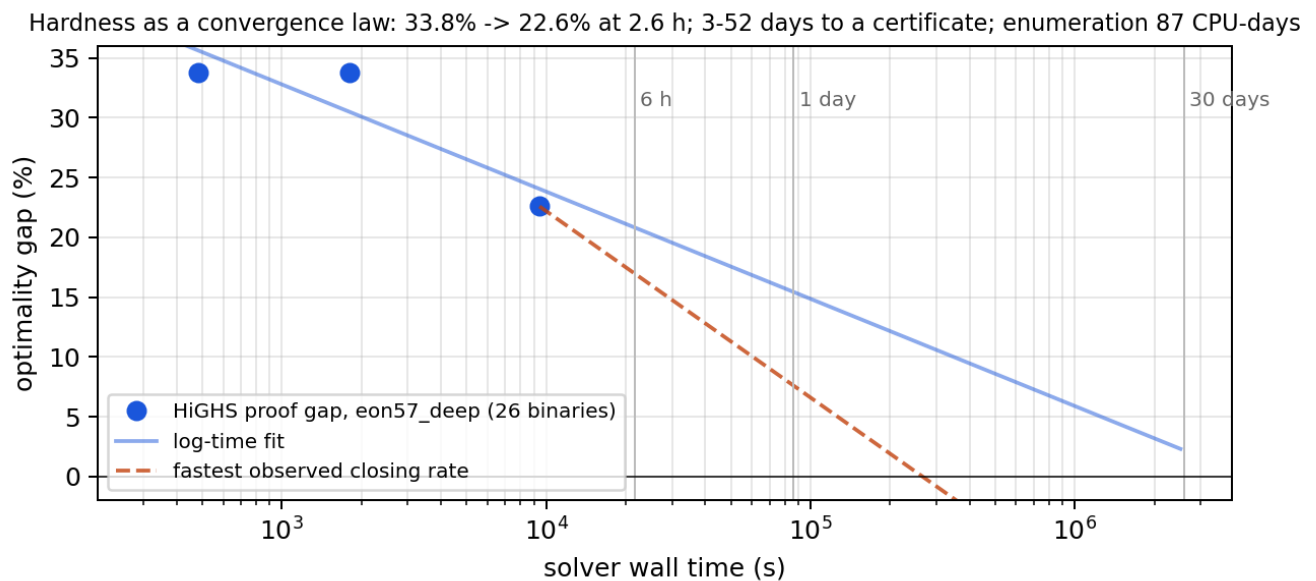


Figure 1 — Hardness as a measured convergence law on eon57_deep: HiGHS optimality gap against wall time, the log-time fit, and the fastest observed closing rate; exhaustive certification would cost 87 CPU-days.

Quantum algorithm on the hard rung. eon57_util on Dirac-3 (degree-2 surrogate over 6,001 referee evaluations, holdout Spearman 0.9993): the device returned in seconds a plan the referee grades at **22,078 — 10.4% better than the 24,652 of the plan the timed-out MILP produced in 901 s**, and better than all 6,001 referee-graded classical samples. A one-flip polish from the device's plan closes the last 0.1% to **22,057**, the best value known for the instance; the same polish from the MILP plan reaches the same value, so the delivered method is the hybrid — device basin-finding in seconds, local polish to the floor — and it is 10.5% better than the proof-grade incumbent at its timeout. Jobs 6a943a6408442f441bbb6ad1 / 6a943a7408442f441bbb6ad2.

Same instance, same referee, all solvers (eon14_hard):

solver	plan	objective	gap
Exact enumeration (2^{10})	certified 10-line set	83.436	—
Dirac-3, native degree-5	same plan	83.436	0.000
QAOA on ibm_marrakesh (d9sqks1dsedc73ai3o30)	same plan	83.436	0.000
Dirac-3, degree-2 (QUBO) restriction	wrong plan	90.995	+9.1%
MPS $\chi=4$ on the true cost	wrong plan (identical to QUBO)	90.995	+9.1%
HiGHS disjunctive TEP	wrong plan	131.87	+58%

MPS gives poor objective value (statement §4.2). Compressing the true 10-variable cost tensor to bond dimension χ and taking its argmin returns the wrong plan at every $\chi < 16$ (+1.9% to +9.1%); the mid-cut operator Schmidt rank saturates its exponential envelope (8/8 at 6 variables, 32/32 at 10). Cheap MPS is exact where the instance is easy

and wrong where it is hard.

Outputs (statement §5.2) — eon14_hard, IEEE 14-bus, four scenarios, N-1: add candidates 0 (4→12), 2 (1→8), 3 (2→6), 5 (4→10), 7 (0→1), 8 (0→4), 9 (1→2); total capex 11.42. Thermal congestion 77.125 → **1.440 (−98.1%)**; overloaded (line, scenario, contingency) triples 227 → 7; voltage-drop violations 423 → 137; exact objective 3,856.27 → 83.436, certified minimum. On 12 held-out scenario draws the certified plan's congestion is 1.229 ± 0.169 vs the MILP plan's 2.068 ± 0.321 (−40.5%).

Hardware on today's devices. ibm_kingston daa0dq6rbfbs73ci56bg (20 qubits, 126 two-qubit gates): certified plan is the best absorber, $\Delta L = +0.113$, ahead of MILP +0.045 and no-build −0.025, clean null and detuned controls. Equal-structure replication on ibm_marrakesh daa9do6rbfbs73cifa80 recovers degree-2 at +0.127 vs frozen +0.124 at visibility 0.93–0.98. Statevector leftover ranks certified > degree-2 > MILP > no-build with principal lines 0.222 / 0.333 / 0.444 cycles per step; the hardware ranking is two-point and partial, and the full-k series is the PoC's first flight.

What it is worth on one decision. Sizing firming storage for one microgrid (20.85 p.u., 24 h, 10% rms) against fluctuation models at identical power coloured by this leftover spectrum: the real-time $S(\omega)$ requires **9.76 ± 0.91 p.u.h**; the two-pole continuation of the same imaginary-time data requires 6.94 ± 0.69 — a **29% undersize** that the real-time route avoids (eon_storage_sizing.json, statevector).

What a successful PoC demonstrates. On E.ON's anonymised MV/LV subgrids: hard rungs certified as such with Gurobi/CPLEX proof gaps; the hybrid device-plus-polish plan graded by the shared referee against the best tuned classical arm at equal total time; and the full-k leftover spectrum measured on hardware for the four candidate plans, feeding storage and reinforcement sizing with a quantity the classical route cannot produce well-posed.

5. Validation plan

Pre-registered per flight: instance, plans, k grid ($k = 0..8$, nine uniform samples, bin spacing 1/9 cycle per step, no FFT padding), shots (8,192 per circuit), layout, estimator, tolerance set against the expected systematic. Controls: no-quench and no-coupling arms for the certified plan (18 of the 54 circuits), detuned arm, two independent runs, second device. Anchors: exact statevector at 20 qubits; the emitter matches the independent matrix model's one-step parities on all four active subspaces. Classical comparators on the same referee with identical scenario access and total time: tuned MILP (HiGHS, Gurobi), multi-start local search from random and from every solver's plan, MPS at reported χ , sensitivity-based screening. Success: (i) hardware four-plan ranking matches the referee order with simultaneous parity confidence bounds inside the pre-registered tolerance; (ii) the hybrid plan is better than or equal to the best classical arm at equal total budget on held-out feeders, or reaches the same feasible objective at lower total cost; (iii) storage sizing from the measured spectrum differs from the continuation route by more than its confidence interval. Falsification: any of the three failing is reported as such.

6. Hybrid / cross-domain integration

The quantum devices are inputs to a classical planning loop that E.ON already runs: candidate generation (Dirac-3, full polynomial) → local polish → referee grading (classical, AC recourse in Phase 2) → contingency-response spectrum (Heron) → storage/reinforcement sizing → plan. Every circuit is a stock Qiskit `QuantumCircuit` run through `qiskit-ibm-runtime SamplerV2` with no custom passes; the Dirac arm ships polynomial files and job IDs for resubmission. Runtime: classical audit ~3 min; each hardware job minutes of QPU time.

7. Team capability

Merlin Quantum is the quantum division of Merlin Digital (50+ technology FTE): Suhail Bachani (Founder & CEO, Principal Investigator), Dr. Hiro Bachani PhD (Program Director), Rohit Bachani (co-founder), Mitul Sawlani (engineering, Purdue), Mohamed Jafun (engineering), Zeena Furtado (finance & operations), Roshan Bhairwani (financial services & deep tech, London), Dr. Ana Baroni MSc (domain specialist). GIC 2026 dual-track finalist. Programme record: 259 receipted QPU jobs, 12.3 million shots, cross-device replication under fixed-layout discipline, and a pre-registration ledger in which failed arms are kept. Closest published analogue is REGRID-QAOA (arXiv:2606.15083); our approach differs in retaining the full degree-5 objective on device, and the measured cost of not doing so is the +9.1% row above.

Appendix A — Hardware job register and receipts

measurement	machine	job id	receipt
eon57_util hard rung: plan graded 22,078 vs incumbent 24,652; polish → 22,057	QCi Dirac-3	6a943a6408442f441bbb6ad1, 6a943a7408442f441bbb6ad2	results/eon57_dirac_result.json, eon57_localsearch_attack.json
eon14_hard native degree-5: certified plan, gap 0.000	QCi Dirac-3	6a79a72508442f441bbb5e56	results/eon_dirac_result.json → native
eon14_hard degree-2 (QUBO) restriction: wrong plan, +9.1%	QCi Dirac-3	6a79a74e08442f441bbb5e57	same → surrogate_deg2
QAOA depth 333, 171 two-qubit gates: certified optimum sampled	ibm_marrakesh	d9sqks1dsedc73ai3o30	results/eon_qaoa_result.json
20-qubit two-point absorption: certified +0.113, MILP +0.045, no-build −0.025	ibm_kingston	daa0dq6rbfbs73ci56bg	eon_field_result.json
Equal-structure replication: degree-2 +0.127 vs frozen +0.124, visibility 0.93–0.98	ibm_marrakesh	daa9do6rbfbs73cifa80	eon_field_v2_result_20260831_004726.json
Instance ladder, HiGHS 0.28 s → 1,801 s; gaps 28.25% / 33.8%	CPU	—	results/eon_ladder.json
Convergence law, 22.6% at 9,437 s; projections	CPU	—	results/eon57_deep_gap_extrapolation.json
Dynamics crossing: lines 0.222 / 0.333 / 0.444; continuation 0.270 misses	CPU (statevector)	—	eon_dynamics_crossing.json
MPS χ -sweep on true cost: $\chi < 16$ wrong plan	CPU	—	results/eon_rubric_mps.json
Storage sizing 9.76 vs 6.94 p.u.h	CPU (statevector)	—	eon_storage_sizing.json
Held-out scenarios −40.5%	CPU	—	eon_v4_advantage.json
Full-k four-plan leftover series (54 circuits)	not flown	—	prepare_application_series.py, series_v6/

Appendix B — Scope notes

The eon57_util device rung used a fitted degree-2 surrogate; both submitted arms returned the identical plan, so that rung establishes improvement over the timed-out incumbent, not an encoding-order effect — the encoding-order evidence is the eon14_hard row (+9.1%). No sister-track 64-rung job is cited as an E.ON result. The 64-rung series is a protocol with the expected signature written before any shot.